

Massage therapy attenuates inflammatory signaling after exercise-induced muscle damage.

Massage therapy is commonly used during physical rehabilitation of skeletal muscle to ameliorate pain and promote recovery from injury. Although there is evidence that massage may relieve pain in injured muscle, how massage affects cellular function remains unknown. To assess the effects of massage, we administered either massage therapy or no treatment to separate quadriceps of 11 young male participants after exercise-induced muscle damage. Muscle biopsies were acquired from the quadriceps (vastus lateralis) at baseline, immediately after 10 min of massage treatment, and after a 2.5-hour period of recovery. We found that massage activated the mechanotransduction signaling pathways focal adhesion kinase (FAK) and extracellular signal-regulated kinase 1/2 (ERK1/2), potentiated mitochondrial biogenesis signaling [nuclear peroxisome proliferator-activated receptor γ coactivator 1 α (PGC-1 α)], and mitigated the rise in nuclear factor κ B (NF κ B) (p65) nuclear accumulation caused by exercise-induced muscle trauma. Moreover, despite having no effect on muscle metabolites (glycogen, lactate), massage attenuated the production of the inflammatory cytokines tumor necrosis factor- α (TNF- α) and interleukin-6 (IL-6) and reduced heat shock protein 27 (HSP27) phosphorylation, thereby mitigating cellular stress resulting from myofiber injury. In summary, when administered to skeletal muscle that has been acutely damaged through exercise, massage therapy appears to be clinically beneficial by reducing inflammation and promoting mitochondrial biogenesis.